

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbially	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 12 (8-12 Nov 2026) - Period 2	Unit 4 - 4.1 Setting the Game Elements (2/2)
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 12 / Lesson 2 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Exit ticket: 70% ordered listener correctly; sessionStorage understood by 80%. Today: loops and asynchronous timing for the game.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students use for/while loops and asynchronous methods (setTimeout, setInterval, clearInterval) to repeat actions and time events. Sprint 4.1 obj. 3-4; Challenges 2-3; CSTA 3A-AP-15.

Key knowledge / skill / understanding:

Loops repeat; 1 s = 1000 ms; asynchronous code runs later, not in order.

Success criteria — what will successful learning look / sound like?

I write a for loop to repeat code; I use setTimeout/setInterval with milliseconds correctly; I stop a timer with clearInterval; my countdown works.

Likely misconception / access or challenge need:

setTimeout pauses the program; loops always need a counter from 0.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Inquiry / investigation ☐ Time: 8 min	Strategy: Problem solving ☐ Time: 20 min	Strategy: Cooperative learning ☐ Time: 8 min
What students will do: Predict-then-run: 4 short code snippets (loop, setTimeout order). Students predict console output, run, then explain differences.	What students will do: Challenge 2: countdown timer from 10 that disappears at 0 and shows a birthday message (or Eid/National Day greeting); then add a loop creating 9 holes dynamically.	What students will do: Pairs swap code and trace each other's timer with a trace table (time, value, output).
Evidence tool: Mini-whiteboards ☐ Assessment: ☒ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group	Evidence tool: Practical / performance task ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Peer-assessment checklist ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group
Resources: Snippets in browser console.	Resources: VS Code; QR 'Asynchronous methods'; source files.	Resources: Trace table template.
Support (low performers / SEN): Annotated snippets; buddy support.	Support (low performers / SEN): Countdown starter with comments; flowchart.	Support (low performers / SEN): Partially filled trace table.
Stretch (fast learners / gifted): Explain why setTimeout(...,0) still runs last.	Stretch (fast learners / gifted): Challenge 3: print primes up to n using loops and if-else.	Stretch (fast learners / gifted): Add an edge case (n = 0 or negative) to test.

3 PREPARE — What do I need to do so students can succeed?				4 RESPOND — What will I do with what I see and hear?			
Set students up to succeed: ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources ☒ Praise, recognition and reinforcement planned				If students struggle or finish early I will: ☒ Re-teach ☒ Regroup ☒ Give extra support ☒ Stretch ☒ Have students redo / improve			
My setup / modelling: Time: 7 min Hook: a countdown on the board - how do games keep time? Model a for loop with a trace table; model setInterval with a counter and clearInterval; convert seconds to ms on the board. Share success criteria.				What I will watch for / likely responses: Infinite loops freezing the tab; forgetting clearInterval; using seconds not ms; counter declared inside the interval. Response: 'my timer goes negative'.			
What is new or innovative in this lesson? <i>(ETEC looks for this)</i> Students choose the celebration message (birthday, Eid, Diwali, National Day) - culturally responsive context; trace tables build computational thinking.				Close / exit check <i>(summative + time for learners' questions)</i> Time: 7 min Exit check: write a setInterval countdown from 5 in pseudocode; convert 2.5 s to ms. Learners' questions.			
Progress recorded / shared: ☒ Gradebook ☐ Parents ☒ Written feedback in notebook / portfolio				Homework / platform <i>(e.g. Madrasati)</i> Madrasati: finish countdown; try the primes challenge (optional). Review student book 4.2 'Start the game'.			

5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>		
What worked — and how do I know? Predict-then-run created real surprise and talk. Celebration choice made it personal and inclusive. 80% built a working countdown.	What will I change next time? Show how to recover from an infinite loop (close tab) before practice. Trace tables need a worked example first.	Students needing follow-up: Students who froze browsers repeatedly (10/4, 10/5) need loop-condition practice. Inform parents of 5 students behind on submissions.

COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i>							
Literacy read + write · 2-1-1-6 ☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Numeracy numbers / data · 2-1-1-6 ☒ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Digital students use it · 2-1-1-4 ☒ 1 ☒ 2 ☐ 3 ☐ C ☒ H	Digital research trusted sources · 2-1-1-4 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Real-life ① application · 2-1-1-5 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Real-life ② application · 2-1-1-5 ☐ 1 ☐ 2 ☐ 3 ☐ C ☒ H	Inquiry experiment / research · 2-1-1-7 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	Thinking problem-solving · 2-1-1-7 ☒ 1 ☒ 2 ☐ 3 ☐ C ☐ H
Create something new · 2-1-1-7 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Discussion expressing ideas · 2-1-1-8 ☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	Pair / group cooperative work · 2-1-1-8 ☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	Learner choice independent work · 3-2-1-6 ☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	Self-assessment 2-2-1-1 ☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	Peer assessment 2-2-1-1 ☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	SEN / low specific support · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H	Gifted specific stretch · 2-1-1-3 ☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback
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